

INVERSE

Modulo n

$$a^{-1} \equiv ?? \pmod{n}$$

Inverses Modulo n

Proposition: $a \in \mathbb{Z}$ is invertible mod n if $\gcd(a, n) = 1$.

proof: $\Rightarrow$ Suppose a is invertible mod n .

There is $b \in \mathbb{Z}$ such that $ab \equiv 1 \pmod{n}$.

$$\Rightarrow n \mid ab - 1 \Rightarrow \exists k \in \mathbb{Z} \quad kn = ab - 1$$

$$\Rightarrow ab - nk = 1 \quad \overset{\text{previous lemma}}{=} d \cdot \gcd(a, n)$$

$$\Rightarrow \gcd(a, n) \mid 1 \Rightarrow \gcd(a, n) = 1.$$

$\Leftarrow$ Suppose $\gcd(a, n) = 1$. So there are $x, y \in \mathbb{Z}$ such that $ax + ny = 1$.

$$\Rightarrow ax = 1 - ny \equiv 1 \pmod{n} \quad \square$$

Teilresultate

Proposition: The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $\gcd(a, n) \mid b$.

proof: $\Rightarrow$: Suppose that x_0 is a solution.

$$\Rightarrow ax_0 \equiv b \pmod{n} \Rightarrow n \mid ax_0 - b$$

$$\Rightarrow ax_0 - b = kn \text{ for some } k \in \mathbb{Z}.$$

$$\text{So } b = ax_0 - nk = ax_0 + (-k)n$$

is a multiple of $\gcd(a, n)$

$$= l \cdot \gcd(a, n) \text{ for } l \in \mathbb{Z}.$$

$$\Rightarrow \gcd(a, n) \mid b$$

$\Leftarrow$: Suppose that $\gcd(a, n) \mid b$. $\Rightarrow b = \gcd(a, n) \cdot m$ for some $m \in \mathbb{Z}$. Write $\gcd(a, n) = ay + nz$ for $y, z \in \mathbb{Z}$.

$$\text{So } b = (ay + nz) \cdot m = a(y_m) + n(z_m).$$

$$\Rightarrow b - a(y_m) = n(z_m) \Rightarrow a(y_m) \equiv b \pmod{n}$$

This means y_m is a solution to $ax \equiv b \pmod{n}$. $\square$

Proposition: Suppose $\gcd(a, n) \mid b$. Then $ax \equiv b \pmod{n}$ has $\gcd(a, n)$ distinct solutions, separated by $\frac{n}{\gcd(a, n)}$.

proof: Set $d = \gcd(a, n)$. Suppose that x_0 is a solution. Consider $x_0 + \frac{n}{d}k$ for $k \in \mathbb{Z}$.

Note: $a(x_0 + \frac{n}{d}k) = ax_0 + (\frac{a}{d}k)n \equiv ax_0 \pmod{n}$
 $\equiv b \pmod{n}$

So $x_0 + \frac{n}{d}k$ are solutions to $ax \equiv b \pmod{n}$

These are separated by $\frac{n}{d} = \frac{n}{\gcd(a, n)}$

$x_0 + \frac{n}{d}k$ is a solution $\forall k \in \mathbb{Z}$.

Question: How many are incongruent.

Suppose $x_0 + \frac{n}{d}k_1 \equiv x_0 + \frac{n}{d}k_2 \pmod{n}$

$$\frac{n}{d}(k_1 - k_2) \equiv 0 \pmod{n}$$

$$\Rightarrow d \mid k_1 - k_2$$

$$\Rightarrow k_1 \equiv k_2 \pmod{\gcd(a, n)}$$

$\Rightarrow$ there are $\gcd(a, n)$ incongruent solutions. $\square$

Strategy for Solving $ax \equiv b \pmod{n}$

① a is invertible modulo n if $\gcd(a, n) = 1$.

$$ax + ny = 1 \quad (x = a^{-1})$$

$$ax \equiv 1 \pmod{n}$$

$$\textcircled{2} \quad ca \equiv cb \pmod{n} \Leftrightarrow a \equiv b \pmod{\left(\frac{n}{\gcd(c, n)}\right)}$$

③ $ax \equiv b \pmod{n}$ has a solution if $\gcd(a, n) \mid b$.

④ If $ax \equiv b \pmod{n}$ has a solution then there are $\frac{n}{\gcd(a, n)}$ solutions separated by $\frac{n}{\gcd(a, n)}$.

Some more properties of $\equiv \pmod{n}$

proposition: If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$

then $a+c \equiv b+d \pmod{n}$

$$a \cdot c \equiv b \cdot d \pmod{n}$$

proof: Suppose $a \equiv b$ and $c \equiv d \pmod{n}$

$$\Rightarrow a - b = kn, \quad c - d = ln$$

$$\Rightarrow a + c - (b + d) = n(k + l)$$

$$\Rightarrow n \mid (a + c) - (b + d) \Leftrightarrow a + c \equiv b + d \pmod{n}$$

$$\text{notice: } a = b + nk, \quad c = d + ln$$

$$\Rightarrow a \cdot c = (b + nk)(d + ln)$$

$$= bd + n(bl + kd + kn)$$

$$\Rightarrow ac - bd = k' \cdot n \Leftrightarrow ac \equiv bd \pmod{n}$$

Question: When can we cancel c in $ca \equiv cb \pmod n$?

proposition: For $a, b, c \in \mathbb{Z}$ we have $ca \equiv cb \pmod n$
 $\Leftrightarrow a \equiv b \pmod{\frac{n}{\gcd(n, c)}}$

proof:

$\Rightarrow$: Suppose $ca \equiv cb \pmod n \Leftrightarrow n \mid c(a-b)$

$\Rightarrow \exists k \in \mathbb{Z}$ mit $c(a-b) = nk$.

Let $d = \gcd(c, n)$. Note $\frac{c}{d}(a-b) = \frac{n}{d}k$.

observe $\gcd(\frac{c}{d}, \frac{n}{d}) = 1$.

$\Rightarrow a-b = \frac{n}{d}k'$ for some $k' \in \mathbb{Z}$

$\Leftrightarrow a \equiv b \pmod{\frac{n}{d}}$

$\Leftarrow$: as exercise. Suppose $a \equiv b \pmod{\frac{n}{\gcd(n, c)}}$

$\Rightarrow a-b = k \cdot \frac{n}{\gcd(n, c)} \quad / \cdot c$

$c(a-b) = k \cdot n \cdot \frac{c}{\gcd(n, c)} = k' \cdot n$

$\Rightarrow ca \equiv cb \pmod n$

□

Some examples $\swarrow$.

examples: $3x \equiv 12 \pmod{15}$

$$\equiv 3 \cdot 4 \pmod{15}$$

$$\Rightarrow x \equiv 4 \pmod{\frac{15}{\gcd(3, 15)}}$$

$$\equiv 4 \pmod{5}$$

$$\Rightarrow x_1 \equiv 4 \pmod{15}$$

$$x_2 \equiv 9 \pmod{15}$$

$$x_3 \equiv 14 \pmod{15}$$

$$6x \equiv 18 \pmod{20}$$

$$\equiv 6 \cdot 3 \pmod{20}$$

$$\Rightarrow x \equiv 3 \pmod{10}$$

$$\Rightarrow x_1 \equiv 3 \pmod{20}$$

$$x_2 \equiv 13 \pmod{20}$$